

Agentic AI Course Syllabus

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Course Overview

This Agentic AI syllabus is designed for learners who want to understand how modern AI systems can reason, plan, use tools, perform multi-step tasks, and support automation workflows. The course moves from basic AI and LLM concepts to agents, tools, memory, planning, orchestration, multi-agent systems, evaluation, security, and real-world implementation.

The focus is practical and project-oriented. Learners will understand the difference between a normal chatbot and an AI agent, how agents interact with tools and data, how to design controlled automation workflows, and how to evaluate agent behavior for reliability and safety.

Prerequisites

Required Foundation

- Basic understanding of Generative AI concepts
- Basic prompt engineering knowledge
- Understanding of LLMs, tokens, context windows, and hallucination
- Basic awareness of embeddings, vector search, and RAG
- Ability to analyze a workflow and break it into steps

For Non-Coding Learners

- Comfort using AI chat tools and browser-based productivity tools
- Basic understanding of business processes such as support, HR, sales, education, or operations
- Ability to write clear instructions and review AI-generated output
- Awareness of privacy, permissions, and approval workflows

For Technical Learners

- Basic programming knowledge in Python, JavaScript, or Java
- HTTP, REST APIs, JSON, and API authentication basics
- Git and GitHub basics
- Basic database knowledge
- Basic backend or web application development
- Optional but useful: Docker, cloud deployment, Spring Boot, Node.js, FastAPI, or LangChain/LangGraph basics

Syllabus Index

Part 1: Foundations of Agentic AI

- Introduction to Agentic AI
- Chatbots vs AI Agents
- Generative AI vs Agentic AI
- Why Agentic AI Matters
- Real-world Use Cases of AI Agents
- Opportunities, Risks, and Limitations

Part 2: Large Language Models for Agents

- LLM Refresher
- Tokens, Context Windows, and Model Behavior
- Reasoning, Planning, and Instruction Following
- Prompt Design for Agent Workflows
- Structured Outputs
- Model Selection for Agentic Systems

Part 3: Agent Architecture

- Core Components of an AI Agent
- Goals, Instructions, and Context
- Planning and Task Decomposition
- Tool Use and Function Calling
- Short-term and Long-term Memory
- Observation, Action, and Reflection Loops
- Agent State Management

Part 4: Tools and Function Calling

- What are Tools in Agentic AI
- Function Calling Concepts
- API-based Tools
- Database Tools
- Search and Retrieval Tools
- File and Document Tools
- Tool Input Validation
- Tool Result Handling

Part 5: Memory in AI Agents

- Need for Memory in Agents
- Conversation Memory
- Task Memory
- User Preference Memory
- Vector-based Memory
- Memory Retrieval and Updating
- Privacy and Data Retention Concerns

Part 6: Planning and Reasoning Workflows

- Task Breakdown
- Step-by-step Planning
- Replanning and Error Recovery
- Human-in-the-loop Review
- Agent Reflection Concepts
- Controlling Long-running Tasks
- When Not to Use Agents

Part 7: Retrieval-Augmented Agents

- RAG Refresher
- Agents with Knowledge Bases
- Document Retrieval for Agent Decisions
- Vector Stores and Semantic Search
- Source-grounded Answers
- Knowledge-base Maintenance
- RAG Agent Project Workflow

Part 8: Multi-Agent Systems

- What is a Multi-Agent System
- Role-based Agents
- Supervisor and Worker Agent Pattern
- Debate and Review Agent Pattern
- Research, Coding, Testing, and Review Agents
- Collaboration and Conflict Resolution
- Risks of Multi-Agent Complexity

Part 9: Agentic Automation

- Workflow Automation with AI Agents
- Email and Communication Automation
- Customer Support Agents
- HR and Recruitment Agents
- Sales and Lead Qualification Agents
- Education and Training Assistants
- Business Process Automation

Part 10: Building Agentic Applications

- Designing Agent Requirements
- Choosing Tools and Data Sources
- Backend Integration Patterns
- Front-end Interface for Agent Workflows
- Logging Agent Steps
- User Approval and Review Screens
- Deployment and Monitoring Basics

Part 11: Agent Evaluation and Testing

- Why Agent Evaluation is Difficult
- Task Success Metrics
- Tool-use Accuracy
- Hallucination and Grounding Checks
- Safety and Policy Checks
- Regression Testing for Agent Workflows
- Human Evaluation and Feedback

Part 12: Security and Responsible Agentic AI

- Prompt Injection
- Tool Abuse and Unsafe Actions
- Data Privacy and Access Control
- Permission Boundaries
- Audit Logs and Traceability
- Human Approval for Sensitive Actions
- Responsible AI Guidelines

Part 13: Frameworks and Ecosystem Overview

- Agent Framework Concepts
- LangChain Overview
- LangGraph Overview
- CrewAI Overview
- AutoGen-style Multi-Agent Concepts
- Spring AI Agentic Patterns Overview
- Choosing a Framework for a Project

Part 14: Capstone Projects

- AI Research Assistant
- Agentic Customer Support Assistant
- Resume Screening and Interview Scheduling Agent
- Sales Lead Qualification Agent
- Document Analysis and Reporting Agent
- Personal Productivity Agent
- Multi-Agent Software Development Assistant

Part 15: Interview Preparation

- Agentic AI Interview Questions
- LLM and Tool-use Questions
- Function Calling Questions
- Memory and RAG Questions
- Multi-Agent System Questions
- Security and Responsible AI Questions
- Project Explanation Practice

Recommended Additions for Industry Readiness

These topics should be included to make the Agentic AI syllabus stronger for production-style learning and real-world automation.

Agent Design Decision-Making

- When to use a simple AI workflow
- When to use an agent
- When to avoid agents and use deterministic software
- Matching agent complexity to business value

Agent Design Patterns

- Single-agent workflow
- Router pattern
- Sequential workflow
- Parallel workflow
- Evaluator-optimizer pattern
- Supervisor-worker pattern
- Human-in-the-loop pattern

Tool Safety and Permission Boundaries

- Read-only tools vs action tools
- Low-risk, medium-risk, and high-risk tool classification
- Approval before sensitive or irreversible actions
- Tool input validation
- Tool result validation

Agent Evaluation

- Task success metrics
- Tool-selection accuracy
- Groundedness and hallucination checks
- Cost and latency evaluation
- Regression testing with real scenarios
- Human feedback loops

Security and Guardrails

- Prompt injection defense
- Sensitive information disclosure prevention
- Excessive agency control
- System prompt leakage awareness
- Unbounded token and cost control
- Audit logging and traceability

Deployment, Monitoring, and Operations

- Agent tracing

- Error recovery and escalation
- Monitoring failed runs
- Versioning tools, prompts, and workflows
- Production-readiness checklist for agents

Detailed Module Plan

Module 1: Foundations of Agentic AI

- Understand what Agentic AI means
- Learn how agents differ from basic chatbots
- Study business and software use cases
- Understand where agents are useful and where they can be risky

Module 2: LLMs for Agentic Systems

- Review LLM behavior, context, and instruction following
- Learn prompt design for goal-based workflows
- Understand structured outputs and model response control
- Study model limitations in multi-step tasks

Module 3: Agent Architecture

- Learn the main building blocks of agents
- Understand goals, context, planning, tools, memory, and state
- Study action-observation loops
- Design a simple single-agent workflow

Module 4: Tools and Function Calling

- Understand how agents call external tools
- Learn API, database, search, document, and file tool patterns
- Practice validating tool inputs and outputs
- Build a tool-enabled assistant prototype

Module 5: Memory and Context Management

- Learn different types of agent memory
- Understand vector-based memory and retrieval
- Study memory update strategies
- Discuss privacy, retention, and user consent

Module 6: Planning, Replanning, and Human Review

- Break complex tasks into smaller steps
- Understand replanning after tool failure
- Add human approval checkpoints
- Build safer workflows for sensitive tasks

Module 7: Retrieval-Augmented Agents

- Combine RAG with agent workflows
- Retrieve trusted context before generating answers
- Use sources to reduce hallucination
- Build a knowledge-base assistant workflow

Module 8: Multi-Agent Systems

- Understand role-based agents and supervisor patterns
- Learn research-review, coding-testing, and planning-execution patterns
- Study coordination challenges
- Design a small multi-agent workflow

Module 9: Agentic Automation Use Cases

- Explore support, sales, HR, education, and productivity agents
- Map business workflows into agent steps
- Identify manual approval points
- Build an automation blueprint

Module 10: Building Agentic Applications

- Convert requirements into an agent design
- Integrate tools, APIs, and data sources
- Add logging and monitoring
- Design a simple UI for reviewing agent actions

Module 11: Evaluation and Testing

- Define success criteria for agent tasks
- Test tool selection and response accuracy
- Evaluate grounding, safety, and reliability
- Create test cases for agent workflows

Module 12: Security and Responsible AI

- Understand prompt injection and unsafe tool execution
- Apply access control and permission boundaries
- Use audit logs and human review
- Prepare responsible AI guidelines for agent projects

Module 13: Projects and Portfolio

- Build guided mini projects
- Document agent workflows on GitHub
- Prepare diagrams for agent architecture
- Present one final capstone project

Suggested Hands-on Assignments

- No. — Assignment — Skills Practiced
- Chatbot vs agent comparison note — Concept clarity
- Tool-calling calculator assistant — Function calling
- API-based weather assistant — API tool integration
- Knowledge-base agent — RAG and retrieval
- Email draft assistant with approval step — Human-in-the-loop workflow
- Task planning assistant — Planning and decomposition
- Multi-agent research workflow — Role-based agents
- Agent safety checklist — Security and responsible AI

Learning Outcomes

By the end of the course, learners will be able to:

- Explain Agentic AI and how it differs from normal chatbots
- Design basic single-agent and multi-agent workflows
- Use tools, function calling, memory, and retrieval in agent systems
- Build simple agentic automation prototypes
- Apply human approval and monitoring for safer agent behavior
- Evaluate agent outputs and tool-use accuracy
- Understand prompt injection, privacy, and permission risks
- Present Agentic AI projects confidently in interviews

What to Learn After Course Completion

Agentic AI Developer Path

- OpenAI Agents SDK
- LangGraph and LangChain
- CrewAI or similar multi-agent frameworks
- Tool calling, tracing, evaluation, and orchestration
- Production agent design patterns

AI Application Developer Path

- Python FastAPI, Node.js, or Java Spring Boot backend
- AI APIs and structured outputs
- RAG and vector databases
- AI-enabled web applications

Java and Spring AI Path

- Spring Boot with Spring AI
- Tool calling and RAG patterns in Java
- PostgreSQL with pgvector
- Enterprise agentic workflow prototypes

AI Automation Path

- Zapier, Make, n8n, and workflow automation tools
- CRM, email, support, HR, and sales automation
- Human approval workflows
- Business process mapping

AI Security and Governance Path

- OWASP Top 10 for LLM and Generative AI Applications
- NIST AI Risk Management Framework
- Google Secure AI Framework
- Agent risk assessment
- Prompt-injection and tool-abuse testing

Recommended Audience

- Students and freshers
- Software developers
- Java and full-stack learners
- AI enthusiasts
- Automation professionals
- Trainers and educators
- Business users exploring AI automation

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